Internal 2 — Stochastic Processes

Maximum Marks: 34 Duration: 90 minutes

Every step must be justified. State clearly any theorem that you use. Part marks are only given for steps that move in the direction of a correct solution.

Notation is as in the class notes. *Advice: Don't solve all problems; you won't have time. The paper is out of 43 marks, and you can drop 9 marks' worth, so do what you know properly.*

1. **(10 marks)** Answer each of the following in one or two precise sentences.

   (a) **(3 marks)** 2 balls are distributed between two urns A and B. At each step one of the 2 balls is selected uniformly at random and transferred to the other urn. Let X_n be the number of balls in urn A after n steps. Write the transition probability matrix of this chain.

   (b) **(3 marks)** If possible, construct a DTMC with one absorbing state and exactly one recurrent non-absorbing state. Explain your answer.

   (c) **(4 marks)** For a DTMC X and states $i, x_0, \ldots, x_n$, prove

   $$\Pr\{X_{n+2} = i \mid X_0 = x_0, \cdots, X_n = x_n\} = \Pr\{X_{n+2} = i \mid X_n = x_n\}$$

   from first principles. Use only the definition of the Markov property and basic theorems from probability theory.

2. **(6 marks)** Let a time-homogeneous DTMC have state space S and n-step transition probabilities $p_{ij}^{(n)}$.

   (a) **(1 marks)** Define what it means for two states $i, j \in S$ to communicate.

   (b) **(5 marks)** Prove that communication is an equivalence relation on S.

3. **(7 marks)** Let $(B_t)_{t\geq 0}$ be a standard Brownian motion and let $(X_t)_{t\geq 0}$ satisfy

   $$dX_t = (2X_t + 3)\,dt + 4\,dB_t, \qquad X_0 = x.$$

   (a) **(4 marks)** Solve the SDE and express X_t in terms of x and an Itô integral.

   (b) **(3 marks)** Compute $\mathbb{E}[X_t^2 \mid X_0 = x]$.

4. **(8 marks)** Let $(S_t)_{t\geq 0}$ satisfy the geometric Brownian motion SDE

   $$dS_t = \mu S_t\,dt + \sigma S_t\,dB_t, \qquad S_0 = s > 0,$$

   where μ and σ are constants. Assume that the solution has the form $S_t = f(t, B_t)$, where f is sufficiently smooth.

(a) **(4 marks)** Use Itô's formula and compare the dt and dB_t terms to show that f satisfies

$$f_t + \frac{1}{2} f_{xx} = \mu f, \qquad f_x = \sigma f, \qquad f(0,0) = s.$$

(b) **(4 marks)** Solve the second equation first and substitute the result into the first equation. Hence find $f(t, B_t)$ explicitly.

5. **(6 marks)** Let $(B_t)_{t \geq 0}$ be a standard Brownian motion. Prove that, for $T \geq 0$,

$$\int_0^T (B_t^3 - 3tB_t)\, dB_t = \frac{1}{4} B_T^4 - \frac{3}{2} T B_T^2 + \frac{3}{4} T^2.$$

You may use Itô's formula.

6. **(6 marks)** A bond is rated good (G), risky (R), or in default (D); default is absorbing. Each year the rating changes according to

$$P = \begin{pmatrix} 0.88 & 0.10 & 0.02 \\ 0.08 & 0.82 & 0.10 \\ 0 & 0 & 1 \end{pmatrix},$$

in the order (G, R, D).

(a) **(2 marks)** Identify the transient and absorbing states. Starting from G and from R, find the probability of default within two years.

(b) **(2 marks)** Find the expected time to default, starting from G.

(c) **(2 marks)** A two-year zero-coupon bond pays Rs. 100 at the end of year 2 if it has not defaulted by then, and Rs. 0 otherwise. With an annual risk-free rate of 5%, compute its present value if it is currently rated G.